Author: Darren Hong Weng Chen

Screenshot of Dashboard

Problem Design

Business Problem

Global Superstore operates across numerous countries, each with distinct levels of economic development, population size, and internet accessibility. While some markets generate strong profits, others may be underperforming relative to their underlying market conditions.

The central challenge for management is determining whether current business performance truly reflects the opportunities available in each market. Identifying countries that have not yet reached their full potential is essential for making informed decisions regarding future investment, expansion, and resource allocation.

Hypothesis

Markets with higher purchasing power, larger populations, and greater internet adoption are generally expected to offer stronger business opportunities and deliver better financial performance.

Consequently, countries with favourable economic conditions but relatively low profitability may indicate untapped growth opportunities that warrant further investigation.

Solution

To address this challenge, an interactive R Shiny dashboard was developed by integrating internal business performance data from the Global Superstore dataset with external economic indicators, including GDP per capita, population size, and internet usage.

The dashboard enables management to compare market potential against actual profitability performance and identify countries that may be underperforming or outperforming relative to their underlying market conditions.

Expected Result

The dashboard is designed to help decision-makers:

- Identify markets with strong growth potential.
- Detect countries that may be underperforming relative to their opportunities.
- Recognise markets that are already performing exceptionally well.
- Support strategic decisions regarding expansion and investment priorities.

By combining business performance data with economic indicators, the dashboard provides a more comprehensive and data-driven approach to evaluating international markets and uncovering future growth opportunities.

Dataset Selection

Primary Dataset: Global Superstore

The primary dataset is the Global Superstore dataset, which contains transactional sales records spanning multiple countries. It captures key business metrics including sales revenue, profit, customer orders, product categories, and geographic locations.

This dataset forms the analytical foundation for evaluating business performance across international markets. The core variables used in this analysis are:

- **Country**: geographic unit of analysis
- **Sales**: total revenue generated per market
- **Profit**: net profitability per market
- **Product Category**: breakdown of performance by product type

External Datasets

Three supplementary datasets were sourced from the World Bank Open Data platform to enrich the analysis with macroeconomic context. All external datasets use 2014 observations to align with the time period covered by the Superstore data.

GDP per Capita

World Bank. *GDP per capita (current US$)*.https://data.worldbank.org/indicator/NY.GDP.PCAP.CD

GDP per capita serves as a proxy for economic prosperity and consumer purchasing power. Markets with higher GDP per capita are generally associated with stronger spending capacity and greater demand for goods and services.

Population

World Bank. *Population, total*. https://data.worldbank.org/indicator/SP.POP.TOTL

Total population provides a measure of market size. Countries with larger populations represent a broader potential customer base and, by extension, greater sales opportunity.

Internet Usage

World Bank. *Individuals using the Internet (% of population)*.
https://data.worldbank.org/indicator/IT.NET.USER.ZS

Internet penetration captures a market's digital connectivity and accessibility. Higher internet usage is typically associated with greater consumer engagement, online commerce activity, and overall business accessibility.

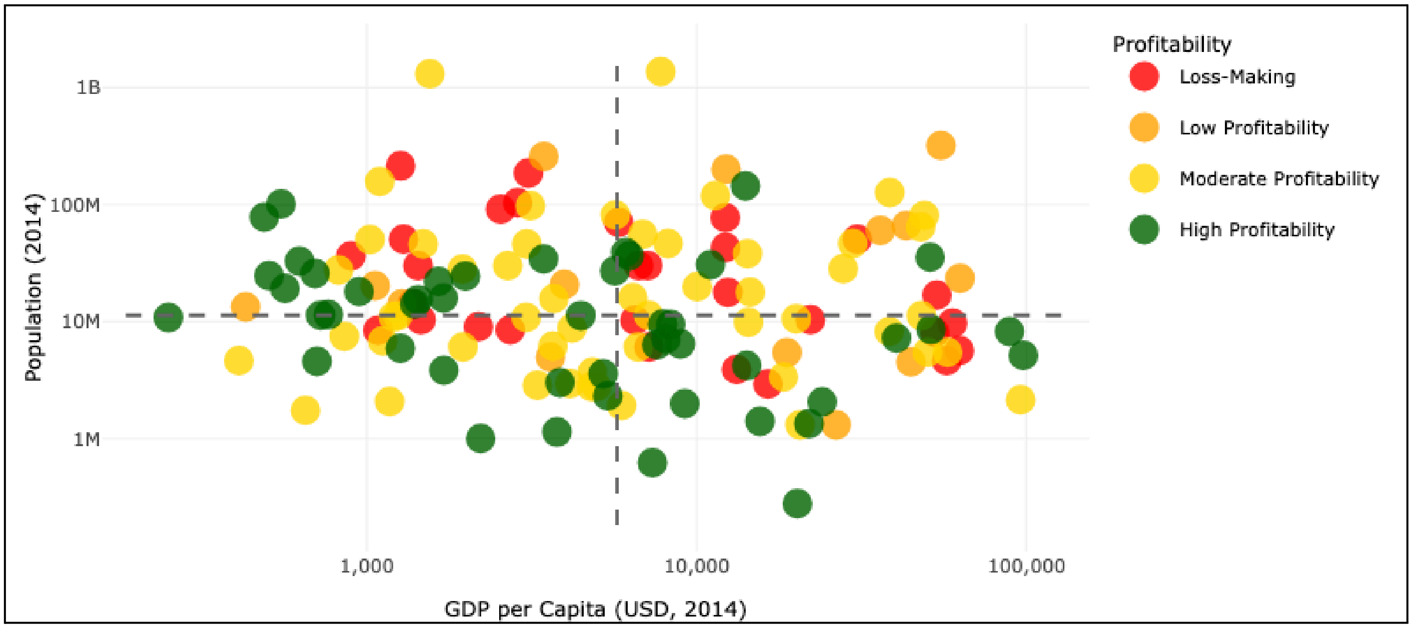
Data Integration

The datasets were integrated by converting country names from the Global Superstore dataset into ISO3 country codes using the `countrycode` package in R. The three external datasets were then joined to the primary dataset using these standardised codes. The resulting integrated dataset pairs internal business performance metrics with external economic indicators, creating a richer analytical basis for evaluating market potential.

Rationale

Relying on profitability alone risks overlooking the structural differences between markets. By incorporating GDP per capita, population size, and internet penetration, the analysis accounts for the economic and demographic conditions that shape what each market can reasonably be expected to deliver. This allows the dashboard to move beyond surface-level comparisons and surface countries that are genuinely underperforming or outperforming; relative to their underlying potential.

Market Potential vs Profitability Performance



This scatterplot maps Global Superstore's performance across international markets by plotting two indicators of market potential: GDP per capita and population size against each country's profitability category, represented by colour.

The dashed reference lines divide the chart into four quadrants, grouping countries by their economic and demographic profile. Countries in the upper-right quadrant combine high purchasing power with large populations, making them structurally the most attractive markets. Those in the lower-left quadrant face more constrained opportunities by comparison.

Three patterns are worth noting:

- **Economic strength does not guarantee profitability.** Several high-GDP countries fall into the low-profitability or loss-making categories, suggesting that favourable conditions alone are not sufficient for strong business performance.
- **Large populations are not being fully converted.** Some of the most populous markets remain unprofitable, which may point to unresolved operational, logistical, or strategic challenges rather than a lack of demand.
- **Similar markets, different outcomes.** Profitability varies considerably among countries sharing similar economic profiles, suggesting that macroeconomic conditions alone do not determine business outcomes. Internal factors such as pricing, competitive positioning, and customer behaviour play a meaningful role in shaping results.

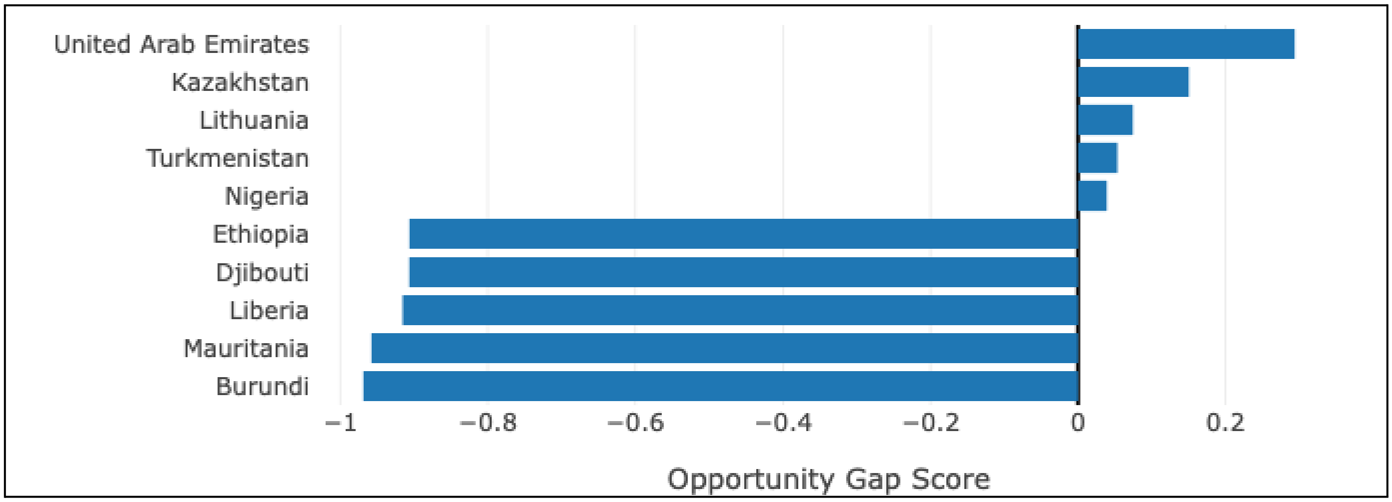
Key Insight

Market potential and profitability are not always aligned. Several countries appear to possess favourable economic conditions but are not translating these advantages into strong profitability, suggesting the existence of untapped growth opportunities that warrant further investigation.

Business Implication

Management should not rely solely on current profit performance when evaluating international markets. Instead, markets that combine strong economic fundamentals with weaker profitability should be prioritised for deeper analysis, as they may offer significant opportunities for future growth and expansion.

Top Underperforming and Overperforming Markets



This visualisation highlights the countries with the largest positive and negative Opportunity Gap scores. The Opportunity Gap metric compares a country's market potential against its actual profitability performance.

A positive Opportunity Gap indicates that a country possesses stronger economic and demographic fundamentals than its current profitability suggests. In other words, these markets may not be fully capitalising on their available opportunities and could represent potential areas for future growth.

Conversely, a negative Opportunity Gap indicates that a country is generating stronger profitability than expected relative to its underlying market conditions. These markets can be considered highly efficient or strategically successful, as they are outperforming what would normally be expected based on their economic characteristics.

The chart therefore separates countries into two important groups:

- **Underperforming Markets (Positive Opportunity Gap)** : Countries with unrealised growth potential.
- **Overperforming Markets (Negative Opportunity Gap)**: Countries that are delivering stronger-than-expected business results.

By displaying both groups simultaneously, the chart allows management to identify where future investments may generate the greatest returns while also recognising markets that could serve as best-practice examples.

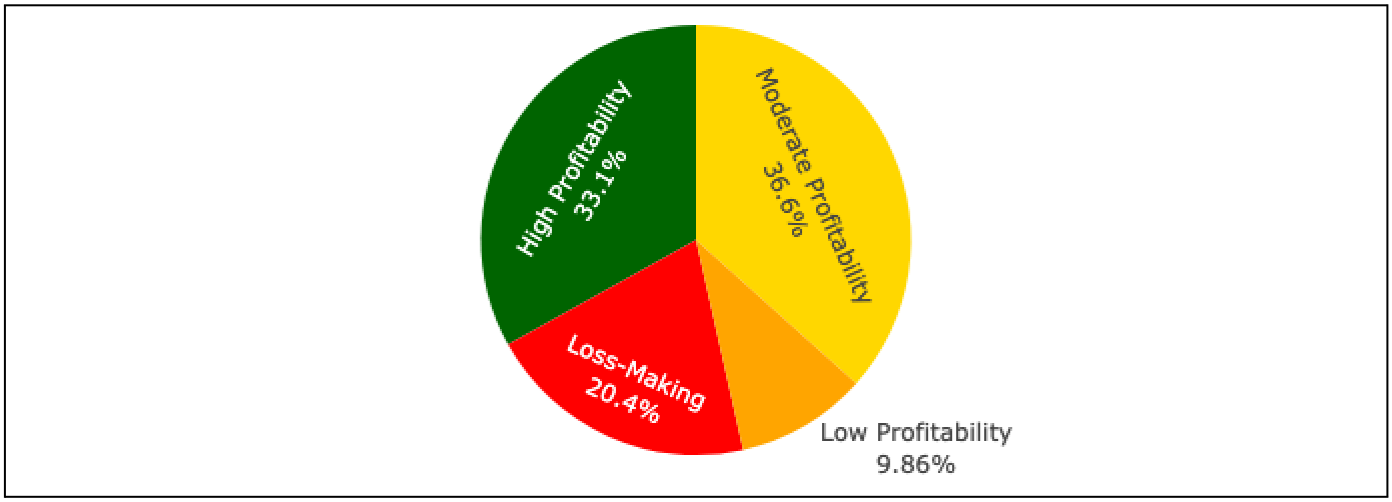
Key Insight

Not all markets perform in line with their economic potential. Several countries exhibit substantial positive Opportunity Gaps, suggesting that favourable economic conditions are not being translated into equivalent profitability outcomes. At the same time, some countries achieve profitability levels that exceed expectations despite having less favourable market fundamentals.

Business Implication

Management should prioritise further investigation into countries with the highest positive Opportunity Gap scores, as these markets may offer the greatest opportunity for future expansion and performance improvement. Meanwhile, countries with strong negative Opportunity Gap scores should be examined to identify successful business practices that could potentially be replicated in underperforming markets.

Market Profitability Composition Pie Chart



This pie chart illustrates the distribution of countries across four profitability categories: Loss-Making, Low Profitability, Moderate Profitability, and High Profitability.

Unlike the previous visualisations that focused on individual markets, this chart provides an overall view of Global Superstore's international market portfolio. By examining the proportion of countries in each category, management can quickly assess whether profitability is concentrated in a small number of markets or distributed more evenly across the business.

The chart shows that countries are not equally distributed across profitability levels. While some markets achieve strong profitability, a considerable proportion of countries remain in the low-profitability or loss-making categories. This suggests that profitable growth opportunities may still exist within the company's current international footprint.

The presence of loss-making and low-profitability markets does not necessarily indicate poor strategic value. Some of these countries may possess strong economic fundamentals but have yet to fully realise their potential, as highlighted by the Opportunity Gap analysis presented earlier.

Key Insight

A significant proportion of markets are not currently operating at high profitability levels. This indicates that there is room for performance improvement across parts of the international portfolio. The profitability composition also suggests that management should avoid focusing solely on the most profitable markets, as opportunities for future growth may exist among lower-performing countries with strong underlying market potential.

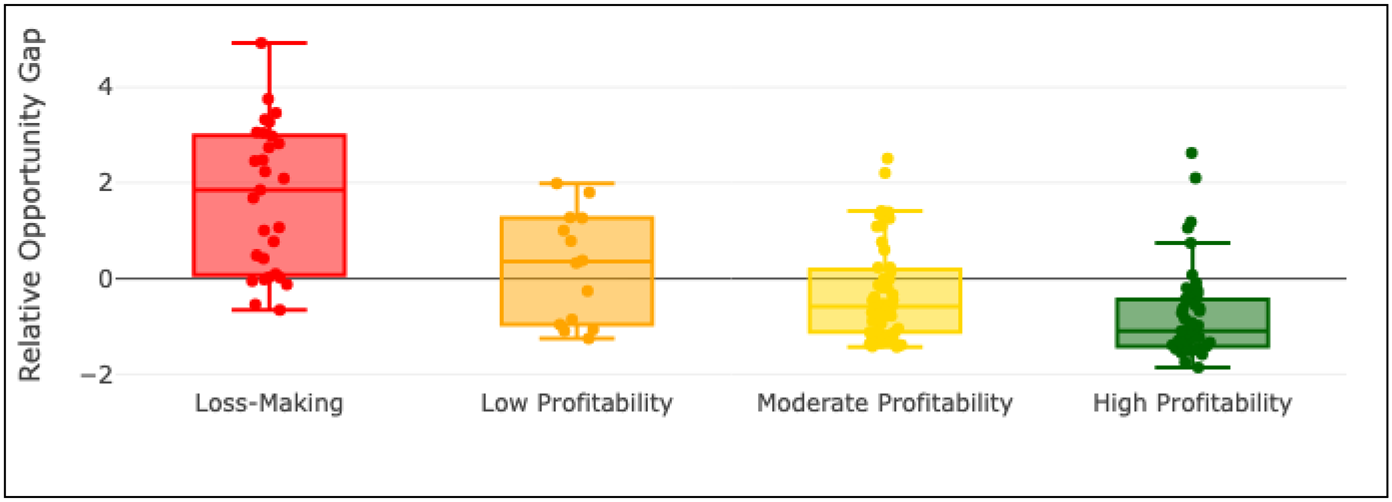
Business Implication

The distribution of profitability categories helps management understand the balance between high-performing and underperforming markets. If a large proportion of countries fall within the low-profitability or loss-making categories, management may consider:

- Investigating operational inefficiencies
- Improving market-specific strategies
- Increasing investment in promising underperforming markets
- Reallocating resources toward higher-growth opportunities

Conversely, markets classified as highly profitable can serve as benchmarks for identifying successful business practices that may be transferable to other regions.

Relative Opportunity Gap by Profitability Category



This boxplot illustrates the distribution of Relative Opportunity Gap scores across the four profitability categories: Loss-Making, Low Profitability, Moderate Profitability, and High Profitability. The Opportunity Gap measures the difference between a country's market potential and its actual profitability performance. Positive values indicate that a market possesses stronger economic fundamentals than its current profitability suggests, while negative values indicate that a market is performing at or above expectations relative to its potential.

A clear pattern can be observed across the profitability categories. Loss-making markets generally exhibit the highest Opportunity Gap scores, with many countries displaying positive values. This suggests that although these markets possess favourable economic and demographic conditions, they are not successfully converting their potential into profitable business outcomes. In contrast, highly profitable markets are concentrated around negative Opportunity Gap values, indicating that they are effectively capitalising on their available opportunities and, in some cases, exceeding expectations.

The spread of the boxplots also provides important insight. Loss-making markets display a wider distribution of Opportunity Gap scores compared to other categories, suggesting greater variability in performance. Some loss-making countries show extremely high positive Opportunity Gaps, indicating substantial unrealised potential, while others perform closer to expectations. Meanwhile, high-profitability markets exhibit a more concentrated distribution, suggesting more consistent performance across countries.

Several outliers are observed across the profitability categories, indicating countries whose Opportunity Gap scores differ significantly from their peers. These markets may represent unique strategic opportunities or challenges that require further analysis.

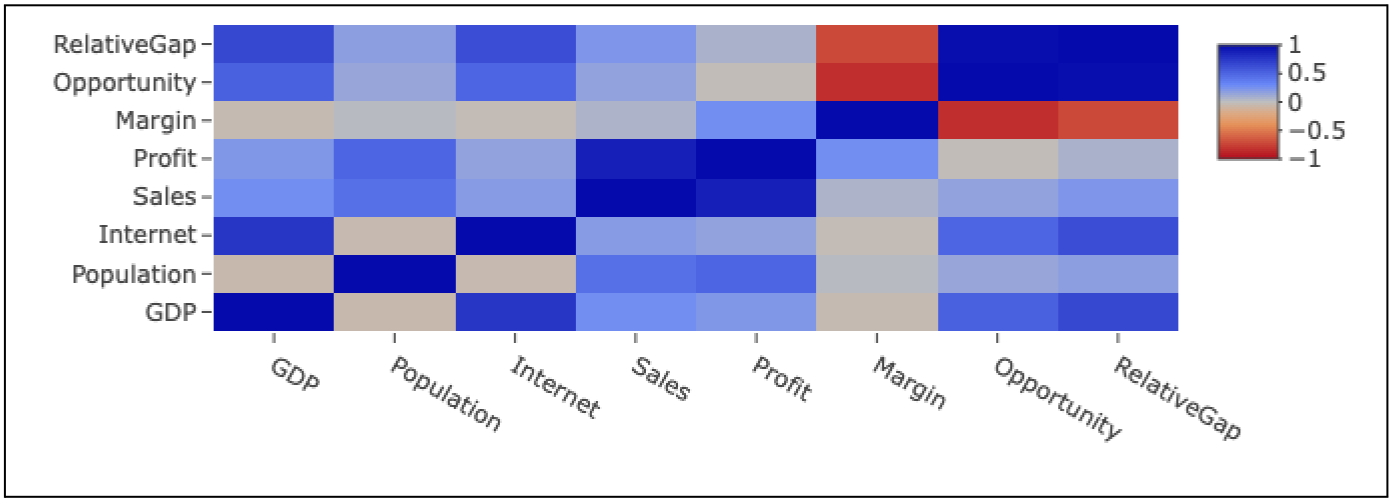
Key Insight

Profitability alone does not fully reflect market performance. Many loss-making and low-profitability markets possess significant untapped potential, while highly profitable markets appear to be extracting substantial value from their underlying market conditions.

Business Implication

Management should avoid making investment decisions based solely on current profitability levels. Countries with large positive Opportunity Gap scores may warrant further investigation, as improvements in pricing strategies, market penetration, customer engagement, or operational efficiency could unlock considerable value. At the same time, highly profitable markets can serve as benchmarks for identifying successful practices that may be replicated across underperforming regions.

Correlation Heatmap of Market and Performance Indicators



This heatmap visualises the strength and direction of relationships between key market indicators and business performance measures. Correlation values range from -1 to +1, where values closer to +1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate little or no relationship.

Several notable patterns emerge from the analysis. GDP per Capita and Internet Usage exhibit a strong positive relationship, suggesting that economically developed countries generally have higher levels of digital connectivity. Total Sales and Total Profit also demonstrate a strong positive correlation, indicating that markets generating higher sales tend to contribute greater profits to the business.

Interestingly, Margin shows relatively weak correlations with GDP per Capita, Population, and Internet Usage. This suggests that favourable market conditions alone do not guarantee profitability. Instead, profitability may be influenced by additional factors such as pricing strategies, operational efficiency, competitive conditions, and customer behaviour.

A strong negative correlation is observed between Margin and Opportunity Gap. This indicates that markets with lower profitability tend to exhibit larger gaps between their potential and realised performance, while highly profitable markets generally operate closer to or above their expected potential. This finding reinforces the usefulness of the Opportunity Gap metric in identifying markets with unrealised growth opportunities.

From a correlation analysis perspective, the heatmap highlights that relationships between variables do not necessarily imply causation. While some indicators move together, they do not automatically cause one another. Nevertheless, identifying these relationships helps management understand which factors are most closely associated with market performance.

Key Insight

Economic and demographic indicators are useful for understanding market potential, but they are not strong predictors of profitability on their own. This reinforces the earlier finding that market potential and business performance are not always aligned.

Business Implication

Management should avoid relying on a single indicator when evaluating international markets. Instead, decisions should consider a combination of economic conditions, market potential, and actual business performance to identify the most attractive opportunities for growth and investment.

Data Story: Unlocking Untapped Market Potential

Global Superstore operates across a diverse range of international markets, each with different levels of economic development, population size, and internet adoption. At first glance, management may be tempted to evaluate markets solely based on current profit performance. However, profitability alone does not provide a complete picture of market attractiveness. A market generating low profits today may still possess significant long-term growth potential, while a profitable market may already be operating close to its maximum opportunity.

The analysis began by examining the relationship between market potential and profitability. The scatterplot revealed that favourable economic conditions do not automatically translate into strong financial performance. Several countries with high GDP per capita, large populations, and strong internet adoption remain classified as low-profitability or loss-making markets. At the same time, countries with similar economic characteristics often produce very different profitability outcomes. This suggests that market success is influenced not only by external economic conditions but also by how effectively the company operates within each market.

To better understand this gap, an Opportunity Gap metric was developed to compare each country's market potential against its actual profitability performance. The results revealed a number of countries with large positive Opportunity Gap scores, indicating that their economic and demographic fundamentals are significantly stronger than their current profitability would suggest. These markets represent unrealised opportunities where potential demand exists, but business performance has not yet fully captured that value.

This finding is particularly important because it challenges the assumption that low-profitability markets should automatically receive less attention. The boxplot analysis showed that many loss-making and low-profitability countries actually possess substantial Opportunity Gap scores. In other words, some of the company's weakest-performing markets may also be some of its most promising growth opportunities. Evaluating markets purely on profitability could therefore lead management to overlook countries that have the greatest capacity for future improvement.

The correlation heatmap provided additional evidence supporting this conclusion. While GDP per capita and internet usage were strongly related, profitability exhibited relatively weak relationships with economic and demographic indicators. This suggests that favourable market conditions create opportunities, but they do not guarantee success. Factors such as pricing strategies, operational efficiency, market penetration, customer engagement, and competitive positioning likely play a significant role in determining whether a market's potential is converted into profitability.

From a strategic perspective, the findings highlight a potential risk for Global Superstore. If high-potential markets continue to underperform, the company may leave valuable revenue opportunities unrealised while competitors strengthen their position in those regions. The cost of inaction is therefore not simply lower profits today, but the possibility of sacrificing future growth and market share in some of the world's most attractive markets.

At the same time, the analysis also identifies opportunities for improvement. Markets that consistently outperform expectations demonstrate that strong profitability can be achieved even when economic conditions are less favourable. These countries can serve as benchmarks for identifying successful business practices that may be replicated in underperforming markets. By learning from these high-performing regions, management can develop more effective strategies for unlocking growth elsewhere.

Overall, the analysis suggests that Global Superstore should move beyond evaluating markets solely on current financial performance. Instead, future investment decisions should consider both market potential and realised profitability. Countries with the largest Opportunity Gap scores should be prioritised for further investigation, as they may offer the greatest opportunities for revenue growth, market expansion, and long-term competitive advantage.

Rather than concluding that some markets are simply unprofitable, the dashboard tells a more compelling story: several of Global Superstore's most promising markets are not yet delivering the results their economic potential would suggest. If these opportunities can be successfully converted into business performance, they may become key drivers of the company's future growth.

AI Acknowledgement and References

Generative Artificial Intelligence (AI) tools were used to assist with brainstorming, code troubleshooting, dashboard design suggestions, report writing refinement, and interpretation improvements. All analyses, visualisations, calculations, business insights, and final decisions were reviewed, validated, and modified by the author before inclusion in this report.

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World Bank Open Data

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Appendix: Processed Dataset Submission

No separate processed or cleaned dataset files are included in this submission.

All data preparation steps, including data cleaning, transformation, integration, and feature engineering, are performed directly within the R Shiny application (`32680120_assignment2.R`). The application imports the original datasets and dynamically generates the processed data structures required for analysis and visualisation at runtime.

Specifically, the application:

- Loads the Global Superstore dataset and external economic datasets.
- Extracts the required 2014 economic indicators.
- Performs data cleaning and missing value filtering.
- Converts country names to ISO3 country codes.
- Joins internal business data with external economic indicators.
- Computes derived variables such as Profit Margin, Potential Score, Profitability Category, and Opportunity Gap.

As a result, no standalone processed dataset was created or stored outside the application. All analytical datasets used by the dashboard are generated programmatically within `32680120_assignment2.R`.